PASS Data Community Summit

Host redgate

Becoming Multi-Platform Proficient and the Tools Which Can Help You

Kellyn Gorman

She/Her

Engineer and Advocate
Redgate Software

I'm the AI and Multi-Platform Advocate at Redgate Software. Along with presenting on open source, AI and Oracle products, I also work with the engineering and product teams on Redgate solutions.

I've been in the tech industry for over 27 years and am known online under my handle, DBAKevlar.

@DBAKevlar

Kellyn.Gorman

https://dbakevlar.com

Gustavo Rene Antunez

He/Him

Principal Solutions Architect

Eclipsys Solutions Inc.

I’m sn innovative Principal Cloud Solution Architect with extensive experience in designing and implementing end-to-end technology projects across multiple industries.

I focus on success in leading cross-functional teams, fostering collaboration to ensure seamless execution of complex cloud architectures. Community-driven and dynamic presenter, with extensive international footprint, recognized for delivering insightful perspectives to diverse audiences in 30+ countries.

@reneace

reneantunez

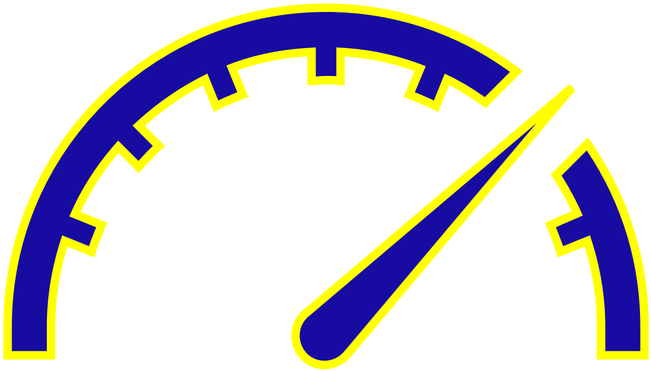
https://rene-ace.com

Agenda

- Why monitoring matters
- Key differences between Oracle, MongoDB, MySQL, and PostgreSQL
- Monitoring vs. Management use cases
- Tooling ecosystem overview
- Proactive alerting and automation
- Real-world examples
- Summary and closing

Why Monitoring Matters

Monitoring isn't just about watching metrics — it's about ensuring performance, stability, and reliability.



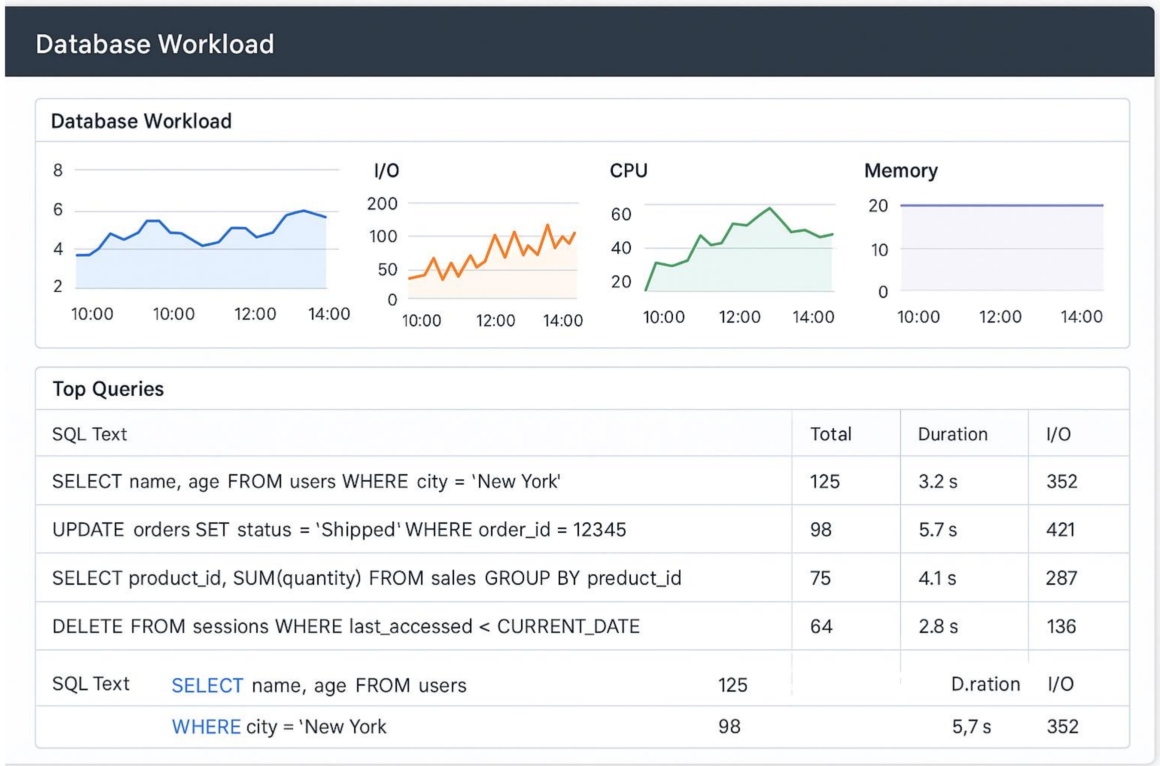
Monitoring Goals

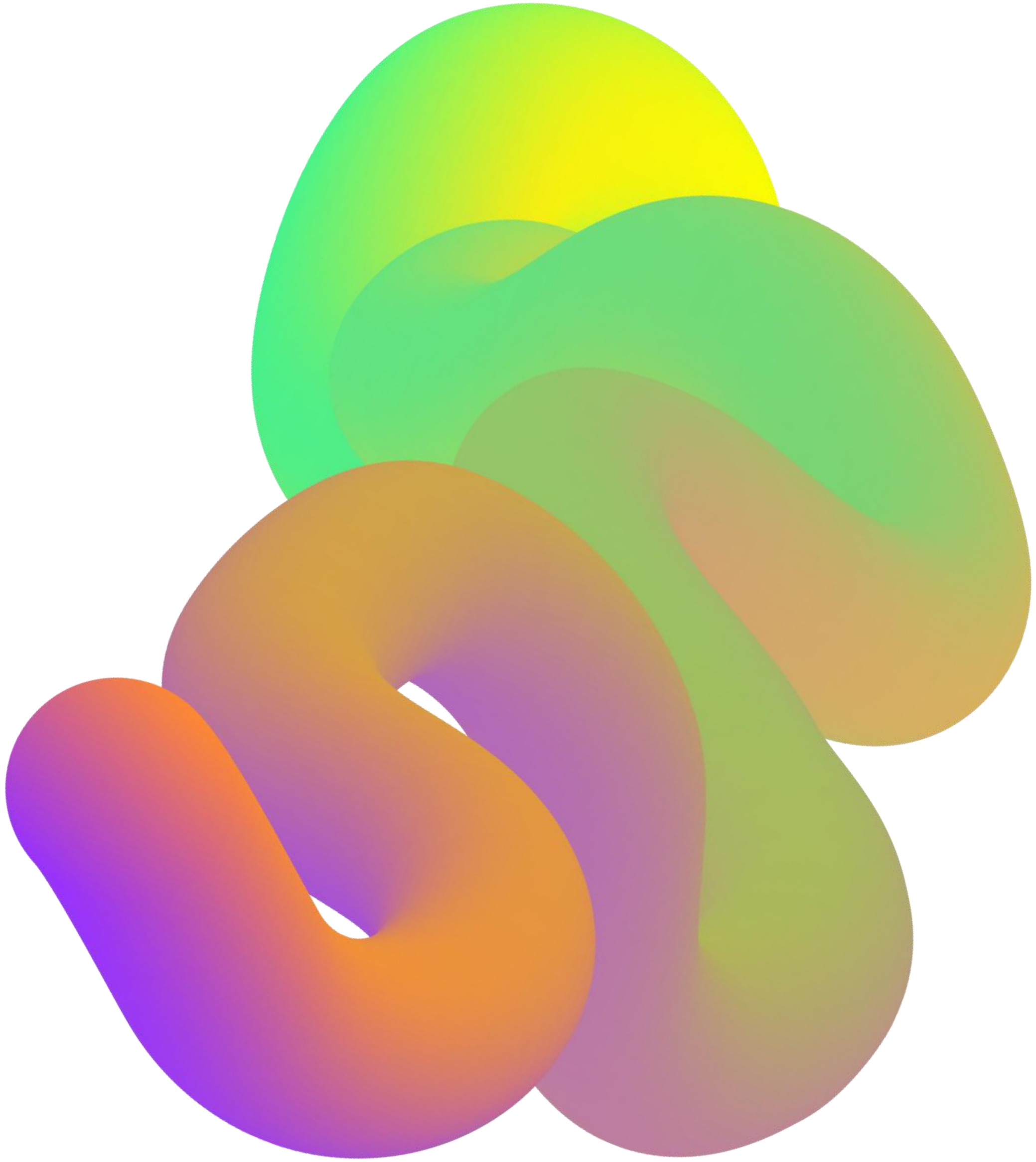
- Detect anomalies early
- Prevent outages
- Identify optimization opportunities
- Ensure compliance and governance
- Support capacity planning

Why Proactive Monitoring Matters

- Detect bottlenecks before users feel impact
- Prevent downtime by identifying trends (i.e., I/O waits, CPU saturation)
- Capture baselines for performance comparison
- Trigger auto-remediation workflows or scaling
- Ensure compliance (SOX, HIPAA, GDPR) through activity and configuration monitoring
- Feed observability data into CI/CD or AI-assisted tuning tools (i.e., Redgate Monitor + Flyway + Ollama AI reviews)

Should Provide Valuable Info



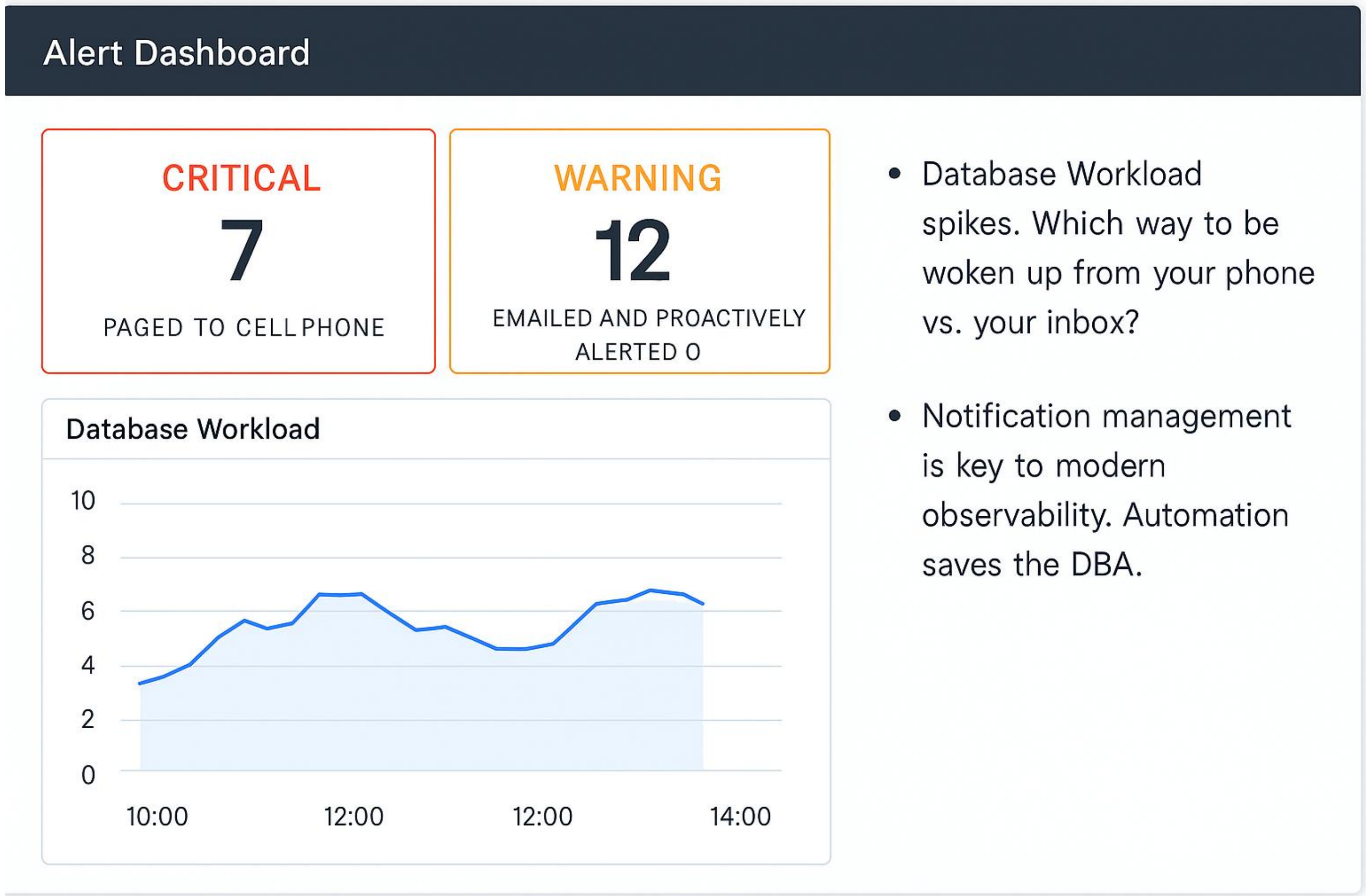


Monitoring vs. Management

Monitoring: Observing, measuring, alerting

Management: Adjusting configurations, applying patches, tuning systems

No White Noise



Notification Example

Critical Alert	Database is down, Database not assessable, Infrastructure outage- Paged Notification 24X7.
High Alert	Application down, but not database you're accountable for. Network issue outside of database- Email notification and Page during workday.
Medium Alert	Storage is at 85% (GBs) full, Temp use is outside of standard levels – email and reviewed weekly.
Low Alert	Storage is 70% full (GBs free) and planning should commence- email and sent to folder for review.
Information Only	Successful deployment to database environment- notified to dashboard only.

Use Cases for Each:

- Monitoring workload: performance, wait events, locks
- Managing database settings: parameters, resource limits
- Managing infrastructure: OS metrics, I/O, CPU, memory usage

Alerting and Proactive Management

Proactive monitoring minimizes downtime.

Why Alerts Matter:

- Prevent issues before users are impacted
- Automate response workflows
- Help teams focus on exceptions, not noise

Examples:

- **ORA-1555 snapshot alerts**
- **MongoDB replication lag thresholds**
- **MySQL deadlock detection**
- **PostgreSQL autovacuum tuning notifications**

Oracle Monitoring

- Oracle Enterprise Manager (OEM)
- Solarwinds Database Performance Analyzer (DPA)
- Redgate Monitor for Oracle
- dbWatch Control Center
- Quest Foglight
- Datadog Database Monitoring
- New Relic Database Monitoring
- And More….

Priorities in Monitoring

- Data Estate/Fleet
- Monitoring
- Database Management(as part of monitoring)
- Multiplatform
- Alerting Complexity
- Ticketing Integration
- Templates/Grouping/Custom Metrics

Oracle Internal Data and Reports

- Automatic Workload Repository (AWR) Reports
- Active Session History (ASH) Reports
- Automatic Diagnostic Database Mgmt. (ADDM) Reports
- Misc. Reporting

AWR Report

…ground Events

Event	Waits	Time(s)	Avg wait (ms)	% DB time	Wait C…
…U		4,424		27.49	
…: TX - row lock contention	3,424	2,467	720	15.33	Application
…g file sync	562,522	1,917	3	11.91	Commit
direct path read	95,418	1,872	20	11.63	User I/O
db file sequential read	495,210	1,774	4	11.02	User I/O

Host CPU (CPUs: 16 Cores: 8 Sockets: 2)

Load Average Begin	Load Average End	%User	%System	%WIO	%Idle
5.98	2.37	8.1	2.2	4.6	88.6

…nstance CPU

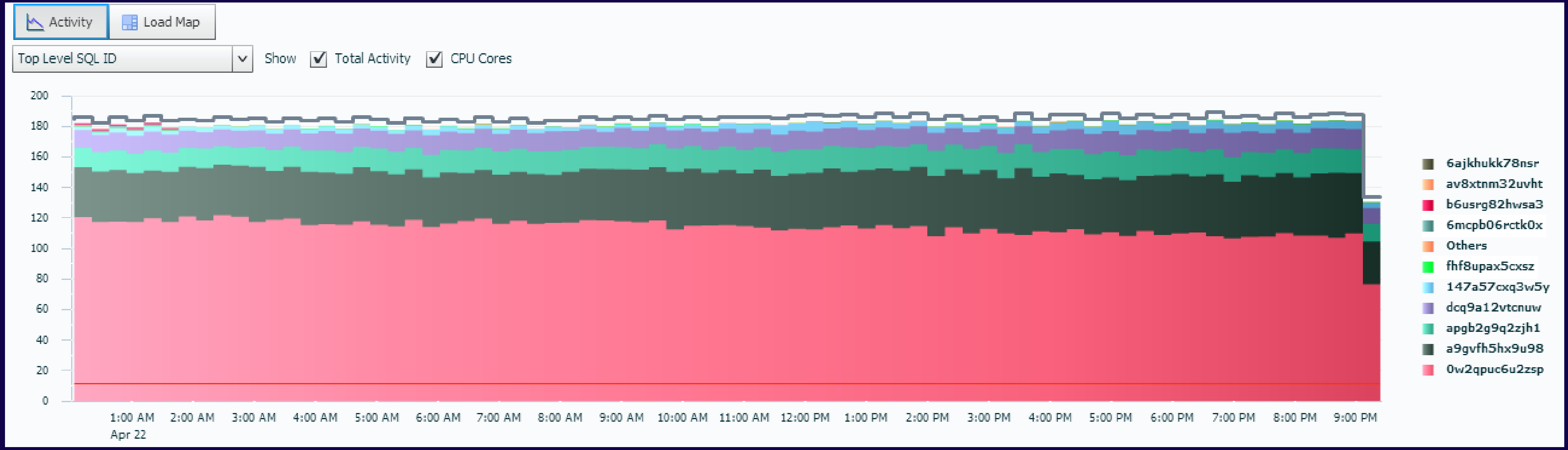
… Total CPU	%Busy CPU	%DB time waiting for CPU (Resource Manager)
9.7	84.9	0.0

Statistics

	Begin	End
	128,845.2	128,845.2
	59,392.0	59…
	2,094.8	

Active Session History

- Taken in Samples
- Overlapping
- Can be run for minute slices of time

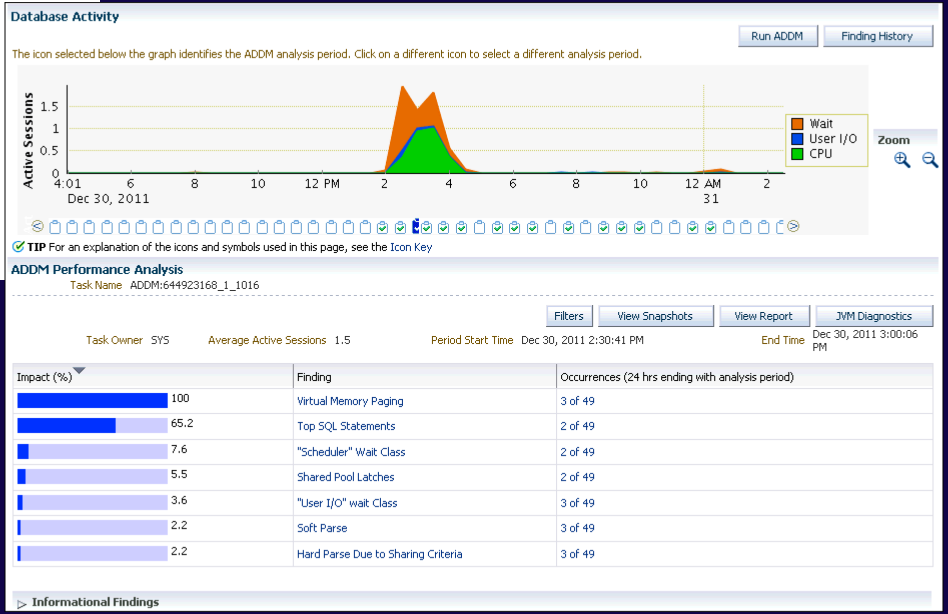


Automatic Diagnostic Database Monitor

```
          Findings and Recommendations
          ----------------------------

Finding 1: "User I/O" wait Class
Impact is 1.76 active sessions, 39.02% of total activity.
--------------------------------------------------------
Wait class "User I/O" was consuming significant database time.
The buffer cache was not undersized.
The streams pool was adequately sized.
The throughput of the I/O subsystem was not significantly lower than expected.

   No recommendations are available.
```



Oracle Management

Management:

- SQL*Plus, Enterprise Manager Cloud Control
- Data Guard Broker for HA management
- OEM or CLI for instance and storage configuration
- SQLcl Project
- Redgate Flyway
- Liquibase

The Needs Decide the Tool

- Managing Database HA/DR
- Migrations/Modernization
- DevOps/Platform Engineering
- Multiplatform Needs
- Development vs. Operations

Oracle is very senior in this space- LOTS of Tools!

MongoDB Tooling

Monitoring:

- MongoDB Cloud Manager / Atlas Monitoring
- MMS and Ops Manager
- Mongostat
- Mongotop
- Quest Foglight for MongoDB
- Redgate Monitor for MongoDB
- LogicMonitor
- Datadog Database Monitoring
- SolarWinds Database Observability

What to Monitor with MongoDB

Metric Category	What to Monitor	Why It Matters
Cluster / Replica Set Health	Node state, optime lag, election statistics	Ensures high availability, consistency and fault tolerance.
Query / Operation Metrics	Reads/writes per second, slow queries, index usage	Helps identify bottlenecks, inefficient queries, capacity issues.
Storage & Cache Usage	WiredTiger cache usage, I/O throughput, working set fits memory	Impacts performance heavily when resources are saturated.
Infrastructure Metrics	CPU, memory, disk I/O, network, connections	Problems upstairs (infra) often masquerade as DB issues.
Alerts & Anomalies	Threshold breaches, unusual replication lag, error rates.	Early warning gives you time to act before user-impact.

MongoDB Management

- mongosh and mongo shell utilities
- Automation via Atlas APIs
- Liquibase
- Redgate Flyway

Focus: Document-level metrics, replica set health, shard balancing, query response times, automation, DevOps

MySQL Monitoring

- MySQL Enterprise Monitor
- Percona Monitoring and Management (PMM)
- SolarWinds Database Performance Monitor
- Datadog Database Monitoring
- Redgate Monitor for MySQL
- Prometheus + Grafana

Key Use Cases for Monitoring MySQL

- **Workload monitoring**: Tools like PMM, MySQL Enterprise Monitor and SolarWinds help you track query response times, slow queries, resource bottlenecks (CPU, I/O).
- **Database settings / tuning**: Many tools provide insights into configuration, indexes, buffer pool usage, etc. (e.g., MySQL Workbench Performance Dashboard)
- **Infrastructure / OS level**: It's important the monitoring tool also covers the host metrics (CPU, memory, disk, network) since MySQL performance often ties to infrastructure.
- **Alerting & proactivity**: Setting up alerts when metrics deviate (latency spikes, replication lag, buffer pool saturation) allows you to act *before* major impact.
- **Historical trends & capacity planning**: Using dashboards that look at performance over time helps you anticipate growth, scale resources, and configure for future load.

Tool	Vendor / Type	Deployment	Key Features & Pros
Percona Monitoring and Management (PMM)	Open-source (Percona)	On-prem / Self-hosted	Deep MySQL query analytics, dashboards, replication lag tracking, free and community supported.
MySQL Enterprise Monitor	Commercial (Oracle)	On-prem / Cloud	Native Oracle tool for real-time performance, query optimization, and security alerts; tightly integrated with MySQL Enterprise.
SolarWinds Database Performance Monitor (DPA)	Commercial	SaaS / Hybrid	Wait-time analytics, query performance baselines, and predictive insights; enterprise-ready with strong visual dashboards.
Datadog Database Monitoring (DBM)	Commercial SaaS	Cloud / Hybrid	Unified app, DB, and infra metrics; real-time dashboards, anomaly detection, and alerting; integrates easily with DevOps workflows.
Prometheus + Grafana (MySQL Exporter)	Open-source	Self-hosted	Custom metrics and visualizations; highly flexible and cost-effective for observability stacks.
Sematext MySQL Monitoring	Commercial / Hybrid	SaaS / On-prem	200+ metrics out of the box, logs + performance data in one view; easy setup with strong alerting.
Redgate Monitor	Commercial	Self-hosted/SaaS	Non-agent MySQL monitoring, thresholds, and alerting; performance data, good host-level visibility.
New Relic Database Monitoring	Commercial SaaS	Cloud / Hybrid	Full-stack correlation of MySQL performance with app and infrastructure; modern UI and quick deployment.

MySQL Monitoring Comparison

MySQL Management

- MySQL Workbench
- CLI tools for tuning and replication configuration
- Redgate Flyway
- Liquibase

Focus: Query response latency, replication lag, InnoDB buffer pool efficiency, automation for deployments

PostgreSQL Tooling

Monitoring:

- pg_stat_activity, pg_stat_statements, pg_stat_all_tables
- OS Level tools: top, iostat, vmstat
- Redgate Monitor/pgNow
- Redgate pgAdmin, pganalyze
- Datadog integrations
- Percona Monitoring and Management
- pgDash
- SolarWinds Database Observability
- Zabbix with PostgreSQL Templates

MySQL and Monitoring/ Management

- Many times, less can be more with MySQL
- Can use a multiplatform tool and it “will do”
- Verify that you can monitor all “flavors” of MySQL. MySQL != MariaDB

Monitoring Tools for Postgres

Tool	Vendor / Type	Highlights / Key Features
Percona Monitoring and Management (PMM)	Open-source (Percona)	Supports PostgreSQL, gives dashboards for PG behaviour, infrastructure usage, query insights.
pgwatch2	Open-source	Lightweight, dockerised PostgreSQL monitoring; configurable, no superuser required.
pgDash	Commercial (but based on open-source collection)	Focused on PostgreSQL only; query analysis, alerts, performance dashboards.
pganalyze	Commercial SaaS	Query optimization, real-time monitoring for PostgreSQL, tuning recommendations.
Prometheus + Grafana (via PostgreSQL exporter)	Open-source stack	Highly flexible, supports system + PostgreSQL metrics, custom dashboards & alerting.
SolarWinds Database Observability	Commercial	Supports PostgreSQL monitoring—latency, throughput, errors, system metrics.
Sematext Monitoring	Commercial / Hybrid	PostgreSQL supported; logs + metrics + alerts.
Zabbix (with PostgreSQL templates)	Open-source	Infrastructure + PostgreSQL metrics, customizable triggers & alerts.

Choosing a Monitoring Tool For Postgres

- **Scope of monitoring**: Do you need just PostgreSQL metrics (queries, locks, replication) or full-stack (DB + OS + infra + application)?
- **Deployment model**: Self-hosted vs SaaS. Open-source vs commercial support.
- **Alerts & proactive monitoring**: Can it trigger alerts on anomalies/thresholds? Does it support baselining and trend detection?
- **Visualization & dashboards**: How good are the dashboards? Can you customize?
- **Query analysis / optimization**: Do you need tools that dig into slow queries, query plans, execution patterns?
- **Cost & operational overhead**: Open-source may be free but require more setup; commercial may be easier but cost more.
- **Integration with existing stack**: Does it integrate with your alerting, logging, DevOps pipelines?

PostgreSQL Management

Tool	Vendor / Type	Key Features & Strengths
pgAdmin	Open-source, official Postgres GUI	Widely used, supports object management, SQL editor, user/role administration.
DataGrip	Commercial (JetBrains)	Advanced SQL editor, cross-database support, refactoring, autocomplete—good for developers working with Postgres + others.
dbForge Studio for PostgreSQL	Commercial GUI/IDE	Visual tools for structure management, query building, data import/export, automation of tasks.
DBeaver	Open-source (with enterprise edition)	Multi-platform, supports Postgres, simple for cross-DB users, good for both development & management tasks. (DronaHQ)
TablePlus	Commercial / Freemium	Modern UI, supports Postgres and other DBs, useful for quick management and data browsing.

Postgres Considerations

- Often multiplatform with enterprise databases.
- Consider needs and future state
- With Greenfield projects, can use multiplatform for a period of time or open-source solutions.

Comparing Platforms Solutions High-Level

Oracle – Deep diagnostic stack with Performance Data and Management Tools.

MongoDB – Cloud-native monitoring and automation APIs

MySQL – Lightweight, flexible open monitoring tools

PostgreSQL – Extensible open-source ecosystem with community-driven tools

Automation and Integration

Integrate alerts with DevOps pipelines:

- Slack, Teams, or PagerDuty notifications
- Custom webhooks for automatic scaling or failover
- DevOps/Platform Engineering automation for schema drift detection

Real-World Example

Scenario:

- A retail company manages multi-database workloads.

Symptoms:

- Latency spikes in PostgreSQL, replication lag in MongoDB

Solution:

- Implemented unified Redgate Monitor dashboard across all systems
- Configured predictive alerts for growth and workload trends
- Reduced downtime by 40% in first quarter

Building a Monitoring Culture

Monitoring isn't a toolset — it's a mindset.

- Treat monitoring as continuous feedback
- Encourage collaboration between DBAs, SREs, and Developers
- Use data from monitoring tools to inform architecture decisions

Management Tools

- Consider the short-term and long-term goals.
- Enterprise solutions vs. open-source- you get what you paid for!
- The more you have to build into the solution, the more human intervention and more human error that may occur.
- Multiplatform is evolving and improving.

Summary

Monitoring ensures reliability, compliance, and insight

Management tools allow proactive optimization

Combine both with automation for intelligent systems

Build a culture that values observability and continuous improvement

Q&A

Thank you!

Rene Antunez | ReneACE | Eclipsys

https://www.linkedin.com/in/reneantunez/

Kellyn Gorman | DBAKevlar | Redgate Software

https://linkedin.com/in/kellyngorman

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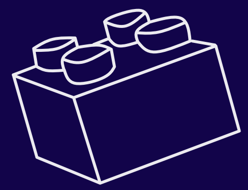

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Come and build some LEGO!

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